



Regulatory Affairs

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May 26, 2023

Docket Nos.: 50-366

NL-23-0409

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant – Unit 2
Licensee Event Report 2023-002-00
Unit 2 Manual SCRAM due to Trip of both Reactor Recirculation Pumps

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(iv)(A), Southern Nuclear Operating Company hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact the Plant Hatch Licensing Manager, Jimmy Collins, at 912.453-2342.

Respectfully submitted,

A handwritten signature in black ink, appearing to read "John Weissinger".

John Weissinger
Vice President – Plant Hatch

HJW/CJC

Enclosure: LER 2023-002-00

Cc: NRC Regional Administrator, Region II
NRR Project Manager – Plant Hatch
NRC Senior Resident Inspector – Plant Hatch
RTYPE: CHA02.004

Edwin I. Hatch Nuclear Plant - Unit 2

Licensee Event Report 2023-002-00

Unit 2 Manual SCRAM due to Trip of both Reactor Recirculation Pumps

Enclosure

LER 2023-002-00



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Edwin I. Hatch Nuclear Plant Unit 2

☒ 050
☐ 052

2. Docket Number

00366

3. Page

1 OF 2

4. Title

Unit 2 Manual SCRAM due to trip of both Reactor Recirculation Pumps

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
03	31	2023	2023	- 002 -	00	05	26	2023	Facility Name	☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

097

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		
☐ OTHER (Specify here, in abstract, or NRC 366A).					

12. Licensee Contact for this LER

Licensee Contact

Jimmy Collins - Licensing Manager

Phone Number (Include area code)

912-453-2342

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
A	AD	81	B093	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On March 31, 2023 at 1430, with Unit 2 in mode 1 at 97 percent power, the reactor was manually tripped due to a loss of both recirculation pumps. The recirculation pumps tripped due to a loss of the 2A and 2B Adjustable Speed Drives (ASDs). All safety equipment responded as expected.

The cause of the ASD trip was due to the high speed stops for the ASDs being set at a value greater than the over frequency relay trip setpoints. This resulted in the relay over frequency trip setpoint being reached prior to the ASD High Speed Stops when increasing ASD speed.

To correct the issue, the ASD High Speed Stop setpoints values were returned to values less than the relay setpoint, to ensure the drive will not be able to be raised to a speed that would result in a trip due to over frequency.

This event is reportable under 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in manual actuation of the reactor protection system.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME		2. DOCKET NUMBER		3. LER NUMBER		
Edwin I. Hatch Nuclear Plant Unit 2		00366		YEAR	SEQUENTIAL NUMBER	REV NO.
☒ 050 ☐ 052				2023	002	00

NARRATIVE**EVENT DESCRIPTION**

On March 31, 2023 at 1430, with Unit 2 in MODE 1 at 97 percent power, the reactor was manually tripped due to a loss of both recirculation pumps [EIS Code: AD]. The recirculation pumps tripped due to a loss of the 2A and 2B Adjustable Speed Drives (ASDs) [EIS Code: AD, SC]. Over frequency relays [EIS Code: AD,81] associated with the ASD's actuated at their trip setpoints based on recirculation pump motor speed.

This event is reportable under 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in manual actuation of the reactor protection system.

EVENT CAUSE ANALYSIS

Causal analysis revealed that the 2A and 2B ASDs tripped due to actuation of over frequency relays when recirculation pump motor speed reached the actuation setpoints. System parameters for the 2A and 2B ASD were reviewed and found that during implementation of a design modification the high speed stop setpoints for the ASD's were set higher than the over frequency relay trip setpoints. The conflicting setpoints is the reason the over frequency trip setpoint was reached prior to the ASD high speed stops and resulted in the ASD trips.

SAFETY ASSESSMENT

There were no actual safety consequences as a result of this event. The Recirculation Pumps tripped as designed to prevent damage due to overspeed. A manual scram was inserted as required by procedures and the reactor protection system worked as expected. Pressure was controlled by EHC and Bypass valves.

CORRECTIVE ACTIONS

To correct the issue the ASD High Speed Stop setpoint values were adjusted lower than the over frequency relay trip setpoint. This change ensures the drive is not able to be raised to a speed that will result in a premature over frequency trip.

PREVIOUS SIMILAR EVENTS

None